

The Involuntary Bending of Reality: Association Between Phenomenological Control and Visual Illusion Sensitivity

Oliver Collins, Ana Neves, & Dominique Makowski



About Me



Studying **BSc Psychology with Clinical Approaches** at Sussex

- Currently on my Placement Year as a Research Assistant at the **Reality Bending Lab**, investigating the neuropsychology of reality perception
- **Joint-winner**: 2025 Jane Beattie Prize for the Best Overall 2nd Year Performance
- **Winner**: 2024 Lisa Rodrigues Prize for the Best Overall 1st Year Performance

Academic interests spanning perception, psychophysiology, research methods, and programming

- Developed four psychology-based Python packages: *PyChroma* (dot-based chromostereopsis illusion generator), *PyGrid* (grid illusion generator), *AfterPy* (afterimage illusion generator), and *Resp-Music* (respiration-controlled music interface)



/OliverACollins



/in/oliver-collins

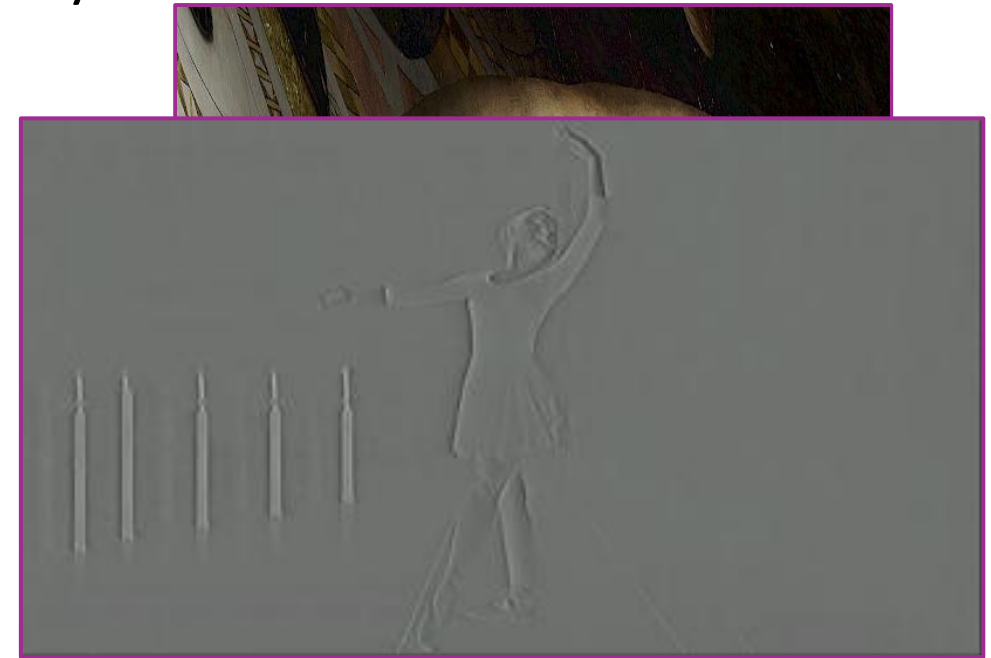


oc236@sussex.ac.uk

Visual Illusions

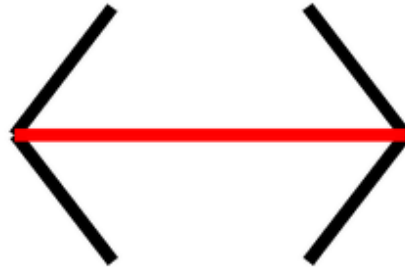
What Are Visual Illusions (VIs)?

- “A **mismatch** between the objective and perceived properties of an object present in the environment” (Notredame et al., 2014)
 - **Conflict** between objective and subjective reality
 - Definitions are fuzzy
- Diversity of VI types
 - But classifications **controversial**
- Long history of interest
 - New illusions **still being created**

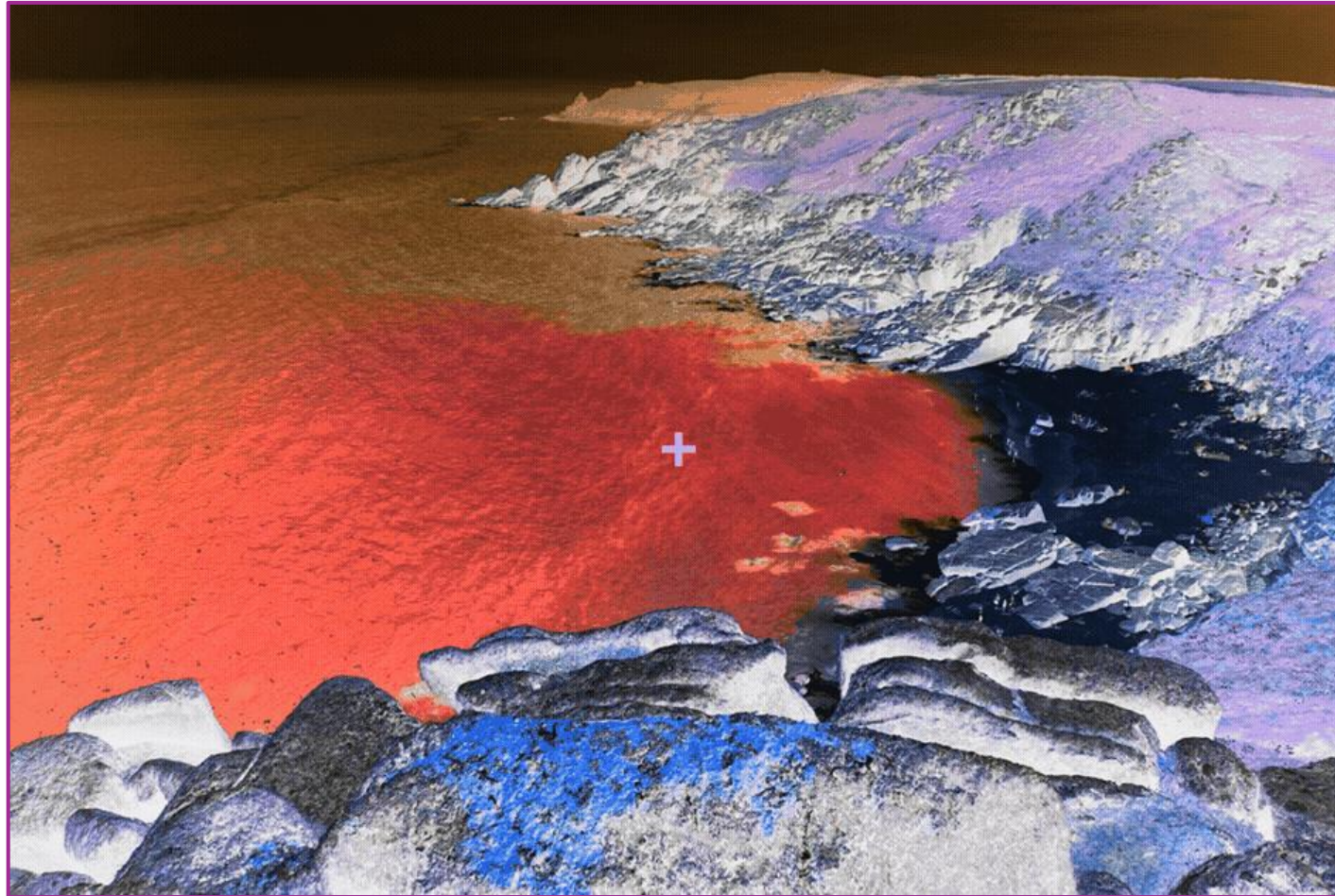


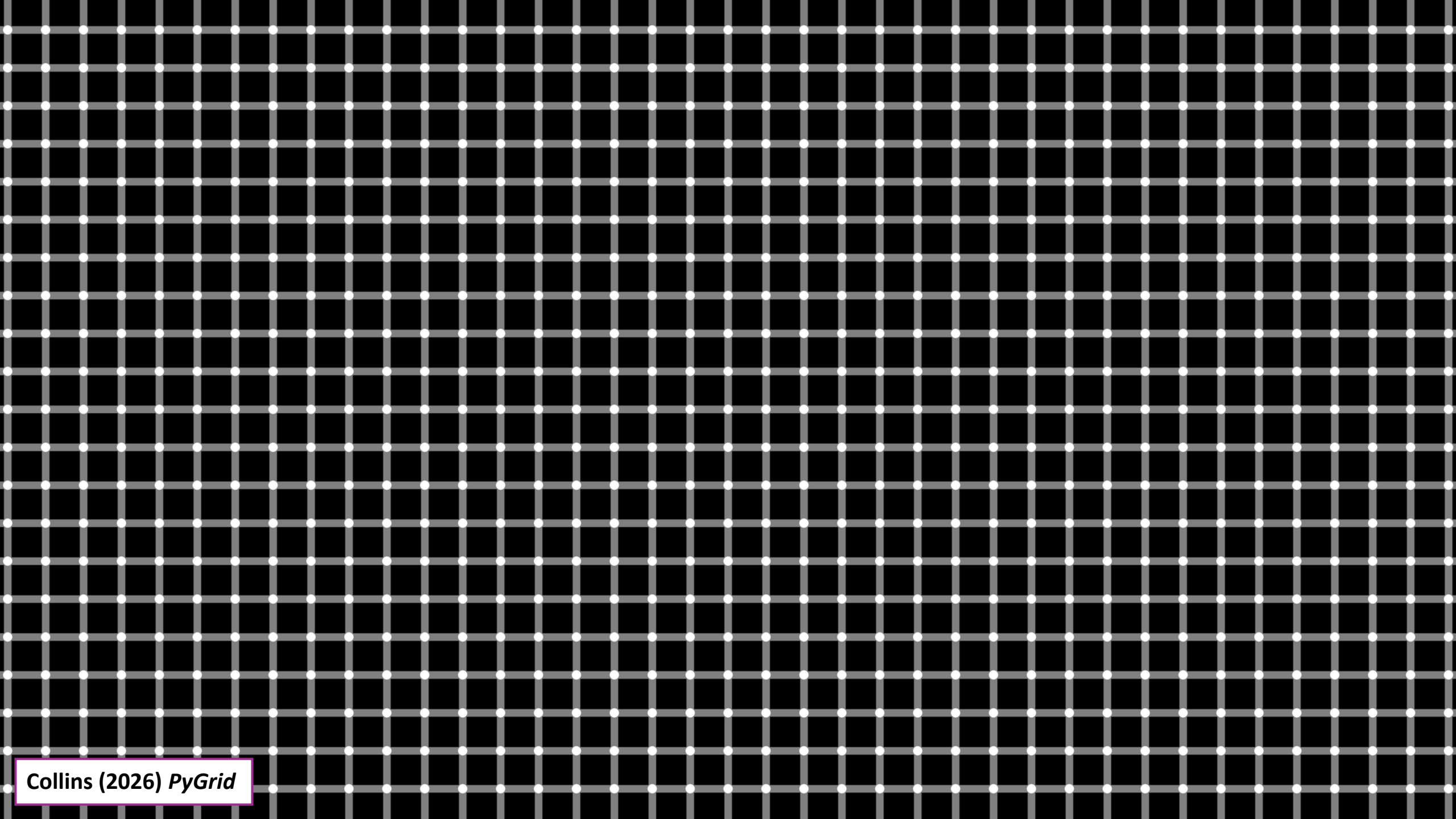
Saleeta Qian and Bernhard Egger, The Static Spin, 2024
Hans Holbein the Younger, 'The Ambassadors', 1533

Examples of VIs



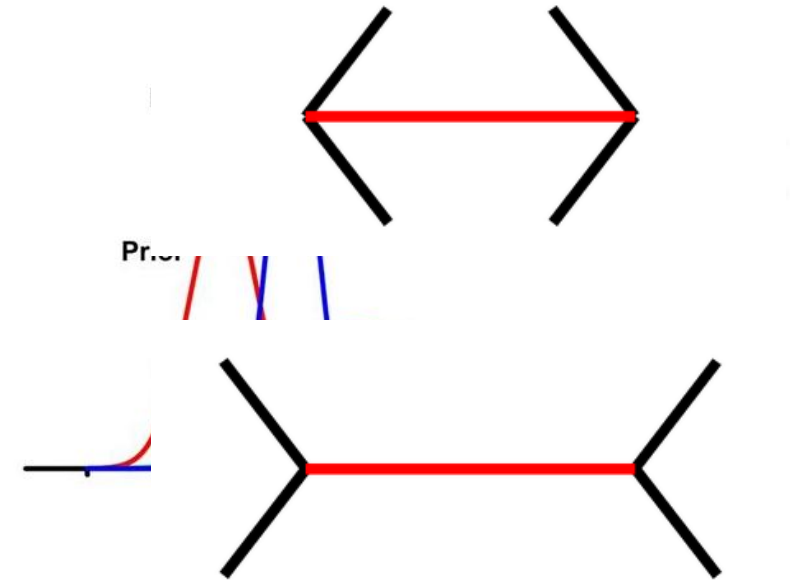
Examples of VIs





What Do VIs Tell Us?

- Canonical example of **Bayesian inference** in perception
- Elucidate fundamental mechanisms underlying the human visual perceptual system (e.g., depth perception, size constancy)
- Potential interindividual differences in perception



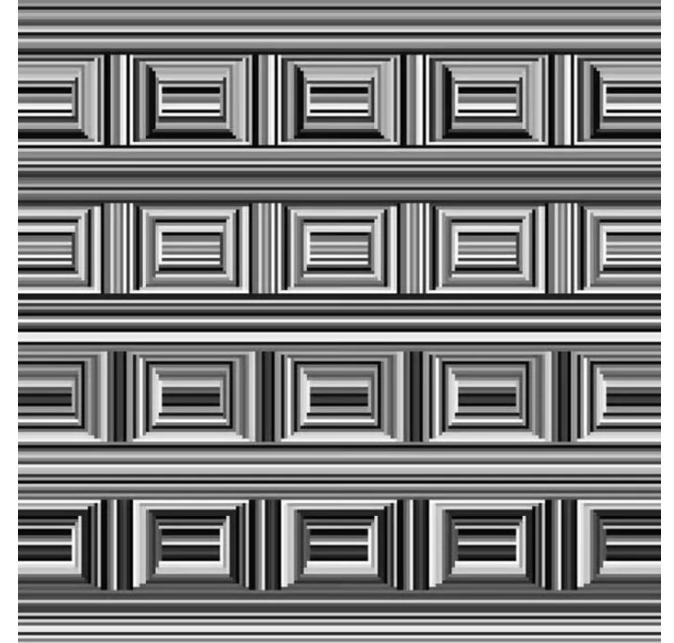
Interindividual Differences in **VI Sensitivity**

Culture

- “Carpenentered-world hypothesis” (e.g., Kroupin et al., 2025)



VS



The Coffer illusion

Interindividual Differences in VI Sensitivity

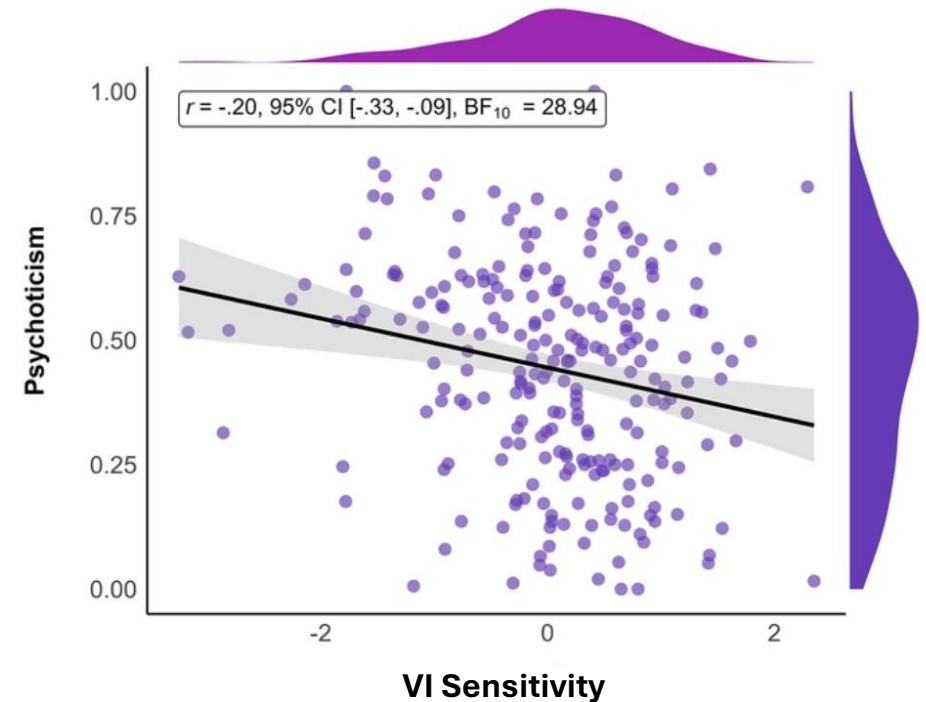
Psychopathology

- Schizophrenia (Costa et al., 2023), autism (Gori et al., 2016)

Personality

- Psychoticism (Makowski et al., 2023)

BUT, mixed findings...



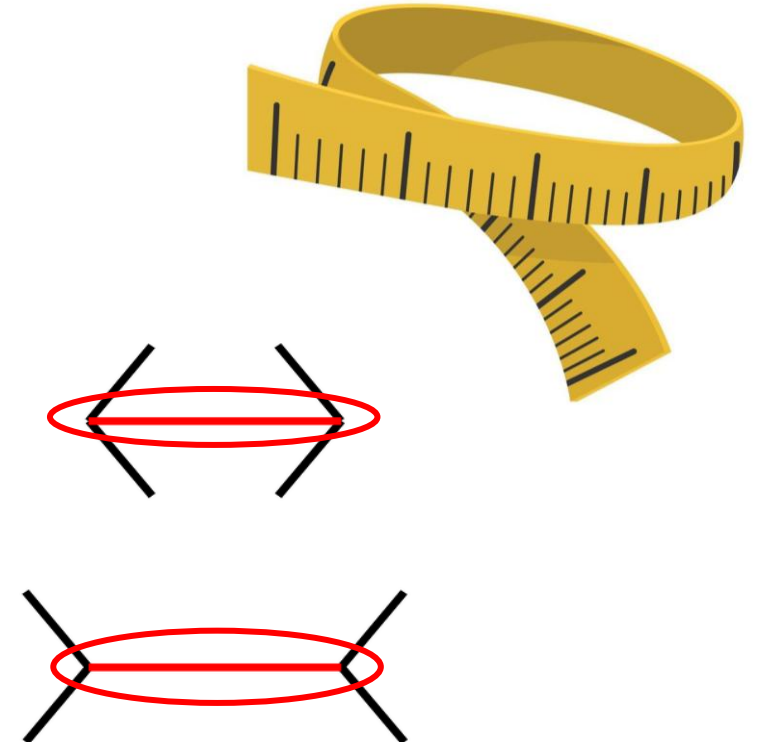
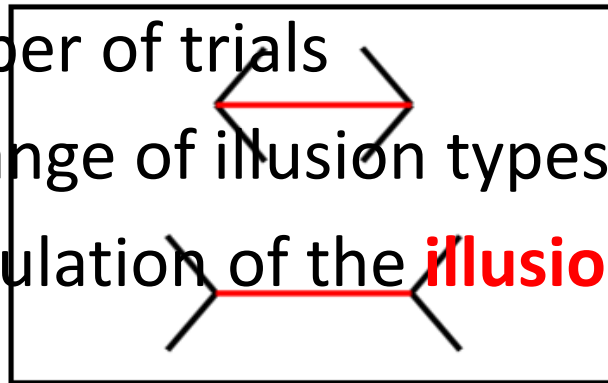
Approaches to Measuring VI Sensitivity

Likert and sliding scales, adjustment tasks, estimation tasks

- Reliance on metacognitive judgements

Use the arrow keys to change the size of the bottom
Typical shortcomings of illusion paradigms

- Low number of trials
- Narrow range of illusion types
- No manipulation of the **illusion context**



Illusion Game

Using Visual Illusions Experimentally

What Is the “Illusion Game”?

- Gamified computerised psychophysics-based 2AFC task aiming to measure **VI sensitivity** (Makowski et al., 2023)
- Participants are presented with different illusion types over 160 trials/illusion
 - Two illusion targets presented simultaneously
- Illusion stimuli generated using the *Pyllusion* package (Makowski et al., 2021)
 - **Parameterised, direct manipulation** of illusions
 - Avoids influence of metacognition
- Indexes **speed-accuracy trade-off**

Py\\usion

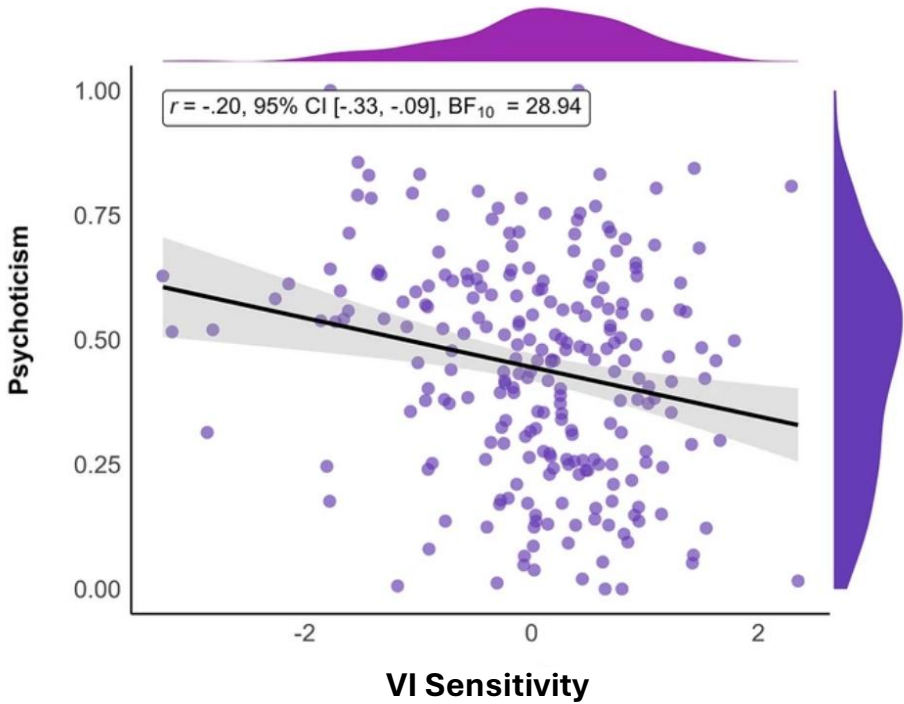
A Parametric Framework to Generate Visual Illusions Using Python

Dominique Makowski ¹, Zen J. Lau², Tam Pham³, W. Paul Boyce ⁴, and S.H. Annabel Chen

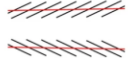
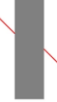
Article | [Open access](#) | Published: 22 April 2023

A novel visual illusion paradigm provides evidence for a general factor of illusion sensitivity and personality correlates

[Dominique Makowski](#) , [An Shu Te](#), [Stephanie Kirk](#), [Ngoi Zi Liang](#) & [S. H. Annabel Chen](#)



Makowski et al. (2023)

Illusion	Example	Task	Description
Delboeuf		Which red circle is bigger? 	Two circles surrounded by outer rings. The circle with the closer outline appears larger than the one with the distant outline.
Ebbinghaus		Which red circle is bigger? 	Two circles surrounded by other circles. The circle surrounded by smaller circles appears larger than the one with the larger surrounding circles.
Müller-Lyer		Which red line is longer? 	Two parallel segments that end with inwards/outwards pointing arrow-like fins. The segment with inward-pointing fins is typically perceived to be longer.
Ponzo Illusion		Which red line is longer? 	Two parallel segments embedded between a pair of converging lines. The segment on the convergence side is typically perceived to be longer.
Vertical-Horizontal		Which red line is longer? 	Two lines segments, one horizontal and one angled. The angled line is usually perceived as longer.
Zöllner		Which direction are the red lines converging towards? 	Represented by two long lines crossed with short, repeated lines, that appear as converging to one side.
Rod and Frame		Which direction is the red line leaning towards? 	A segment ("rod") enclosed in a tilted square ("frame"), that appears to be leaning to one side.
Poggendorff		Is the right line above or below the left line? 	Two oblique segments separated by a rectangle. The right segment appears to be vertically misplaced.
Simultaneous Contrast		Which small rectangle is lighter? 	Consists of two smaller grey rectangles embedded in backgrounds of different contrasts. The rectangle with the darker background appears lighter.
White		Which vertical rectangle is lighter? 	Composed of a series of contrasting horizontal bars superimposed with vertical grey rectangles. Rectangles superimposed onto darker bars appear lighter.

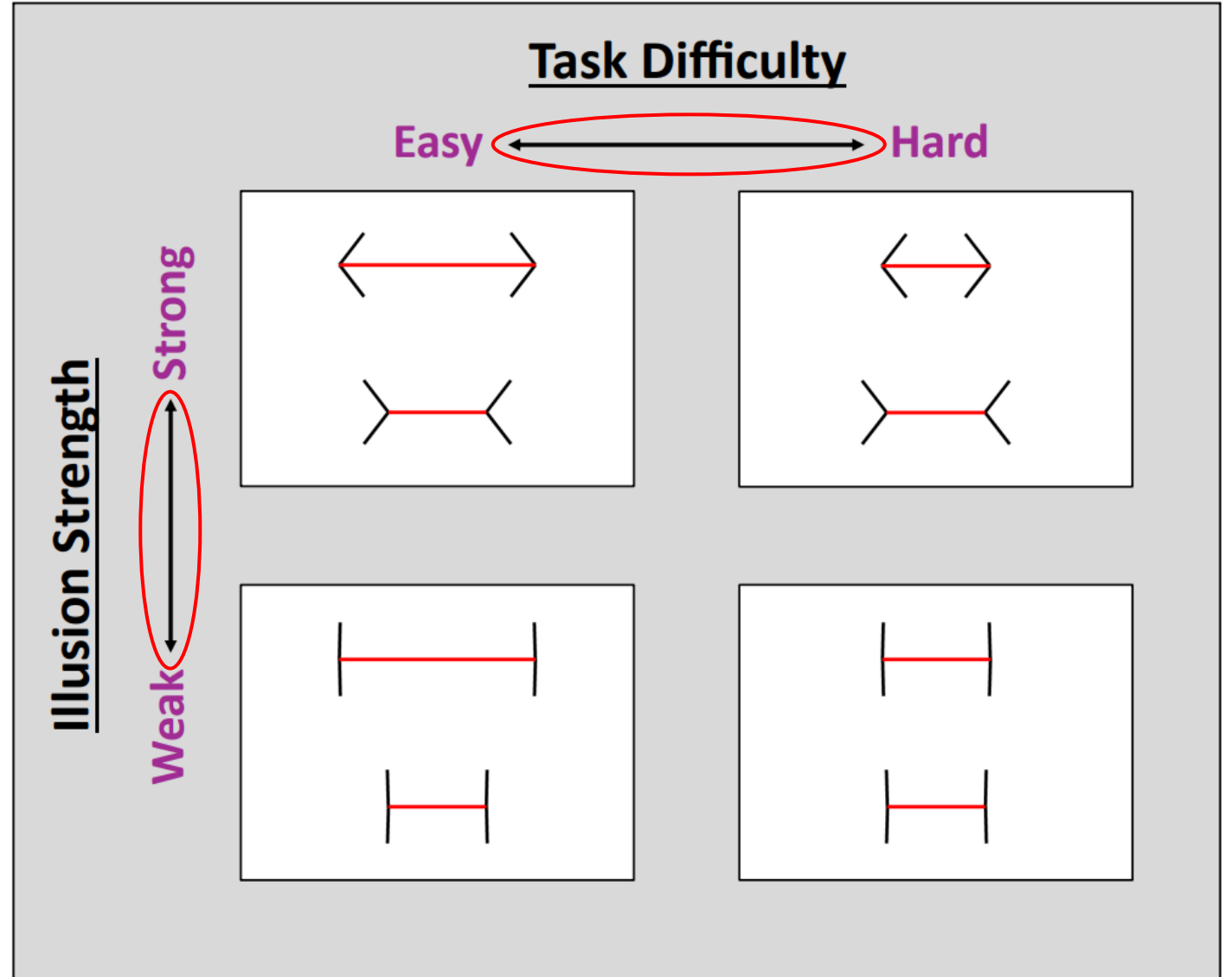
Parameters of the Illusion Game

Task Difficulty

- The relative objective size difference between **core illusion elements**

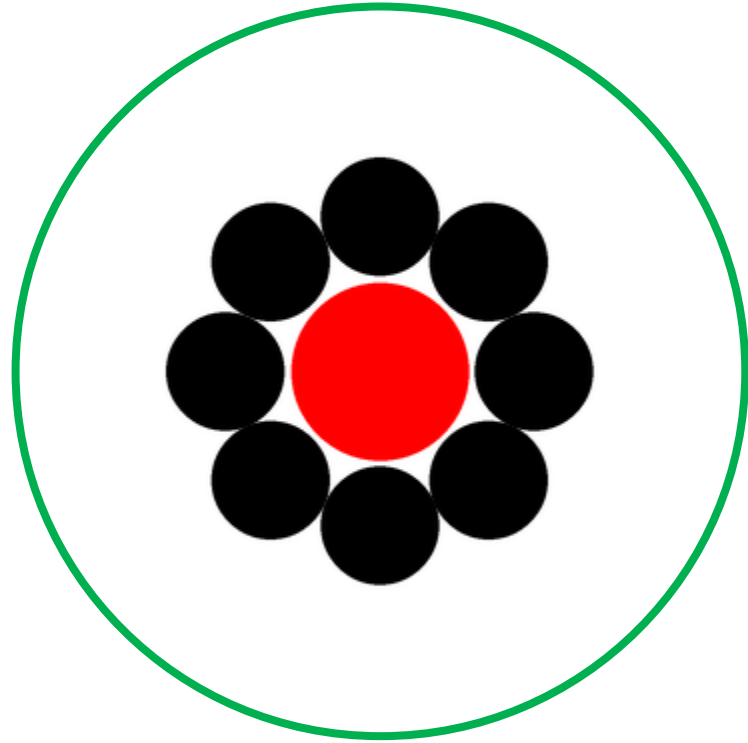
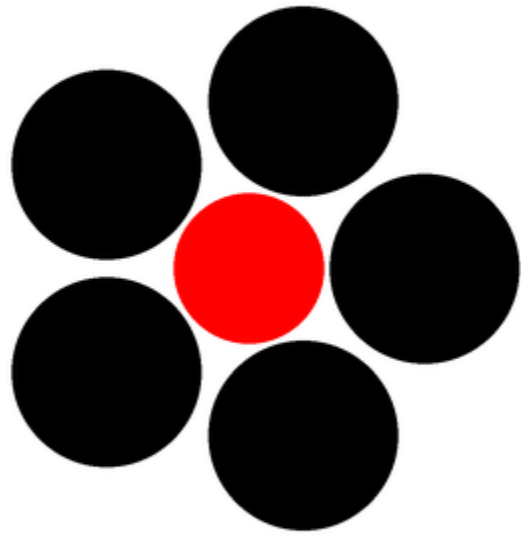
Illusion Strength

- The extent of perceptual interference from other illusion elements (**the context**)

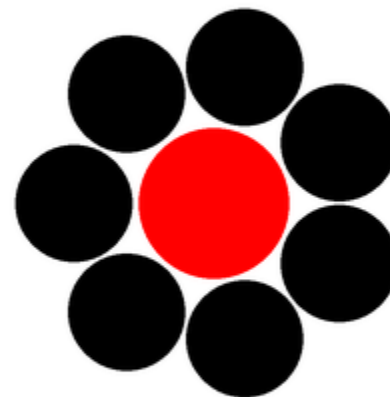
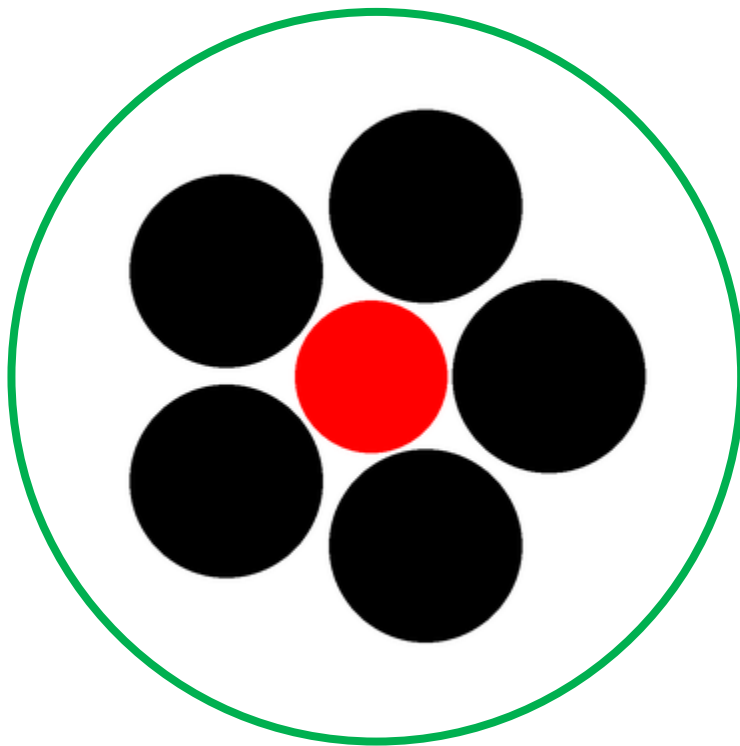


Let's play the Illusion Game!

Which red circle is bigger?

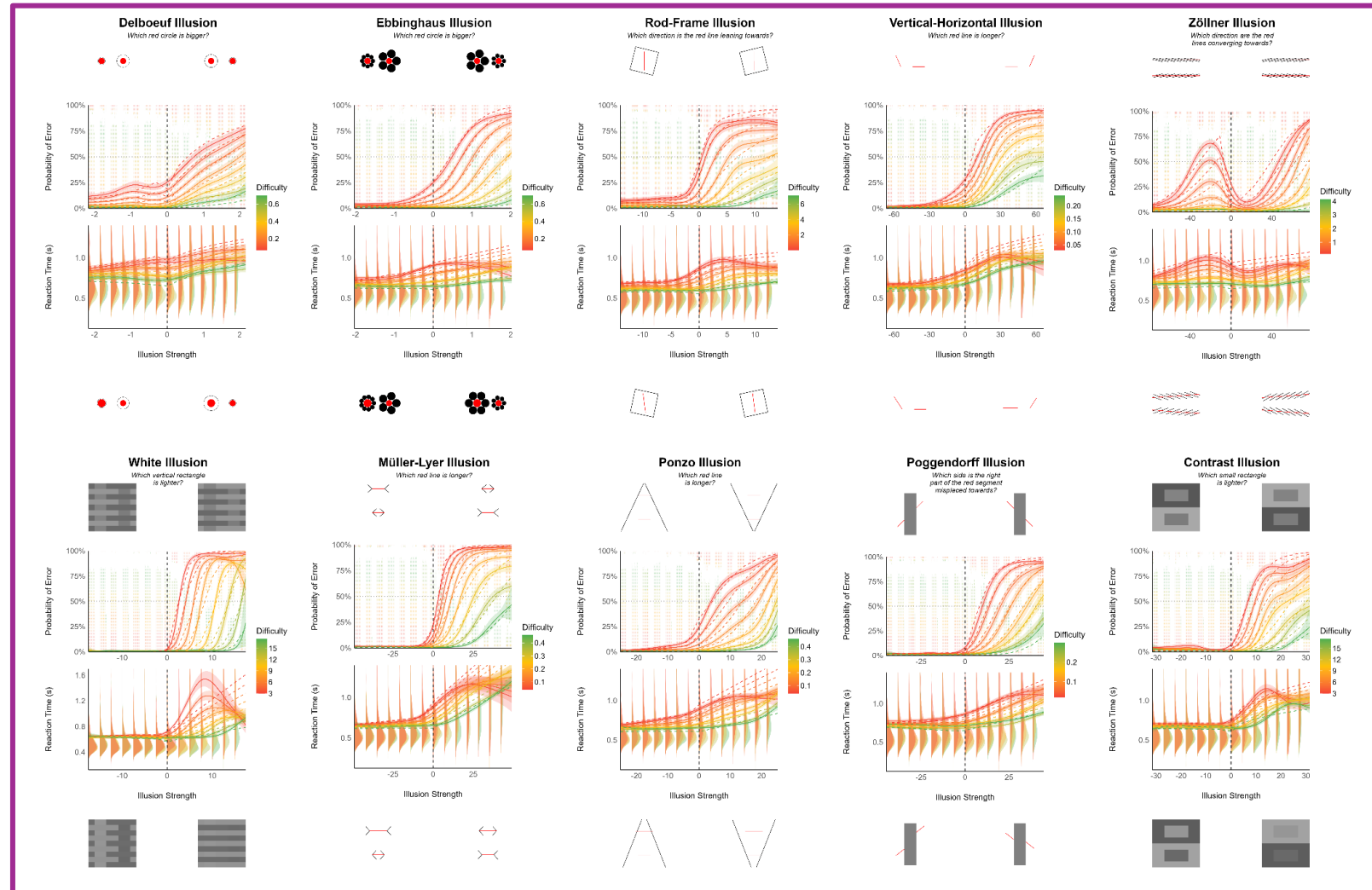


Which red circle is bigger?



Quantifying VI Sensitivity

Generalised Additive Models (GAMs)



Interindividual Differences in VI Sensitivity

Psychopathology

- Schizophrenia (Costa et al., 2023), autism (Gori et al., 2016)

Personality

- Psychoticism (Makowski et al., 2023)

Phenomenological control?

Visual Illusions and Phenomenological Control

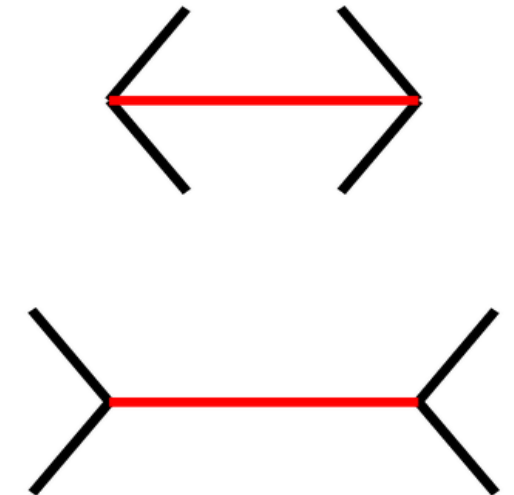
- VIs and PC both involve **altered perceptual experiences**
- Is VI sensitivity related to one's ability to "**resist**" an illusion stimulus?
 - Task-demand characteristics in experimental setting

Lush et al. (2022)

Trait phenomenological control in top-down and bottom-up effects: ASMR,
Visually Evoked Auditory Response and the Müller-Lyer illusion

Lush, P.*^{1,2}, Dienes, Z.^{1,3}, Seth, A.K.^{1,2,4}, & Scott, R. B.^{1,3}.

- Explored the relationship between **PCS scores** and sensitivity to the **Müller-Lyer illusion** within a “line discrimination task”
 - Canonical example of a “cognitively impenetrable” VI
 - **Cognitive impenetrability hypothesis** (Pylyshyn, 1999): basic/early visual perception is not under conscious control through higher-level cognitive processes
 - Perceptual judgement task, 50 trials
 - Evaluating comparison and reference targets
 - Manipulation of **task difficulty**
- Found an **absence** of relationship between PC and VI sensitivity



Our Study

Hypotheses

- Replication and extension
- Stage 2 registered report

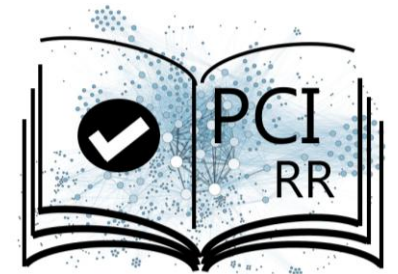
Testing the Relationship between Phenomenological Control related to Illusion Sensitivity

Dominique Makowski and Ana Neves

School of Psychology, University of Sussex

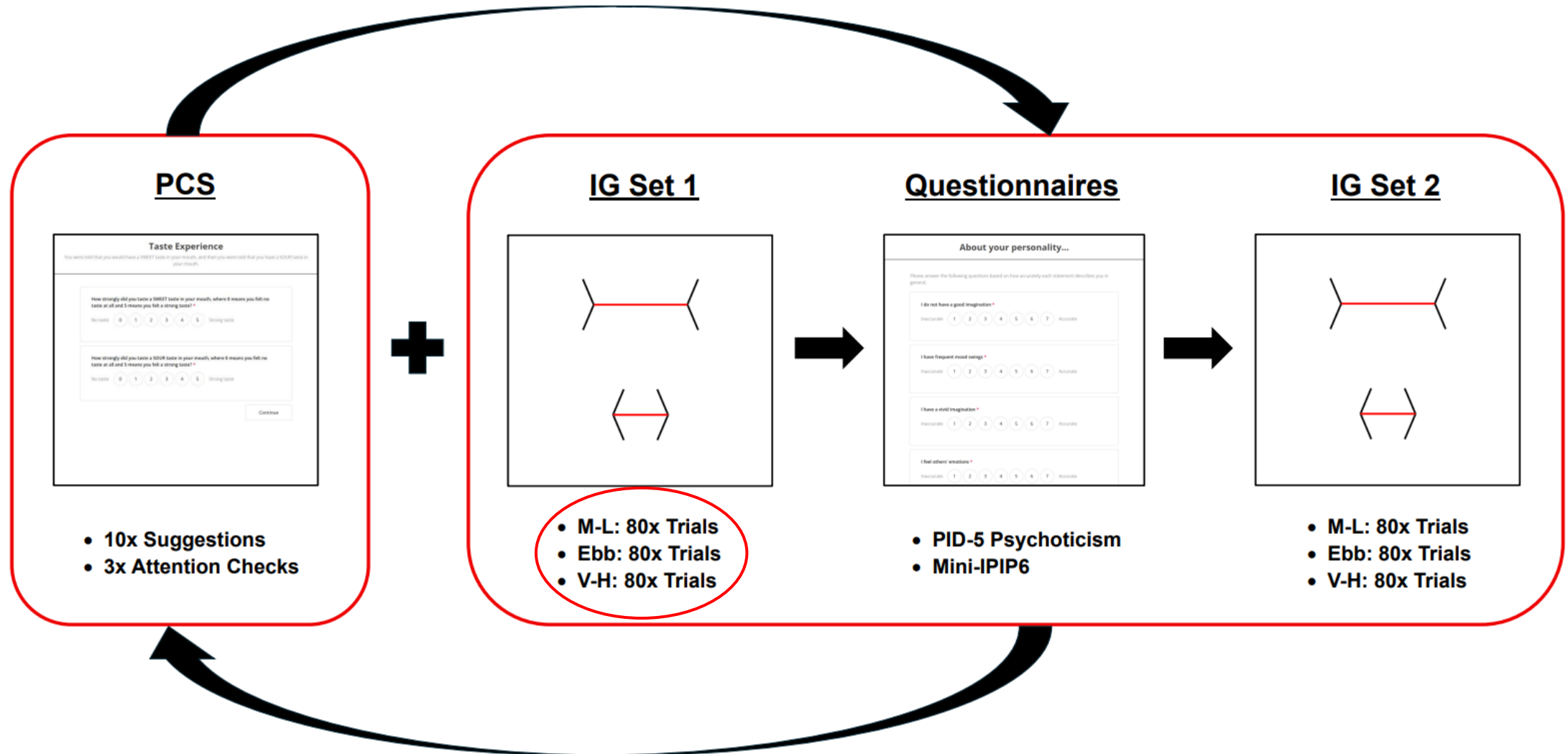
(1) Based on Lush et al. (2022), it was predicted that there would be evidence supporting the **absence** of correlation between **phenomenological control** and **visual illusion sensitivity** ($BF_{10} \leq 1/3$)

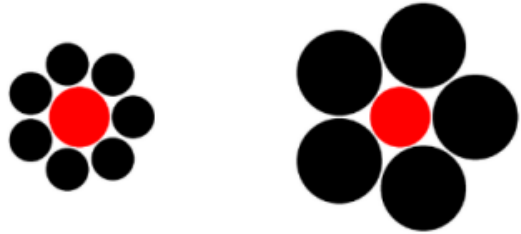
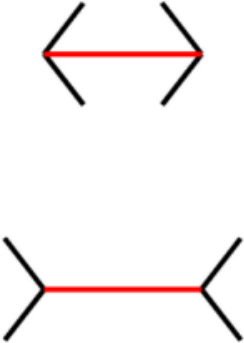
(2) Based on Makowski et al. (2023), it was predicted that **psychoticism** scores would be **negatively correlated** with **visual illusion sensitivity** ($BF_{10} > 3$)



Methods

- Online data collection via **Prolific** ($N = 500$)
- Final sample: $N = 464$ ($M_{\text{Age}} = 38.6$ years, $SD = 13.1$, range = 18–75, 50% ♀)
- jsPsych adaptation of the **Phenomenological Control Scale** (PCS; Lush et al., 2021) and the **Illusion Game** (IG; Makowski et al., 2023)



Illusion Type	Example	Task
Ebbinghaus		<i>Which red circle is bigger?</i>  
Müller-Lyer		<i>Which red line is longer?</i>  
Vertical-Horizontal		<i>Which red line is longer?</i>  

Quantifying VI Sensitivity

Inverse Efficiency Score (IES)

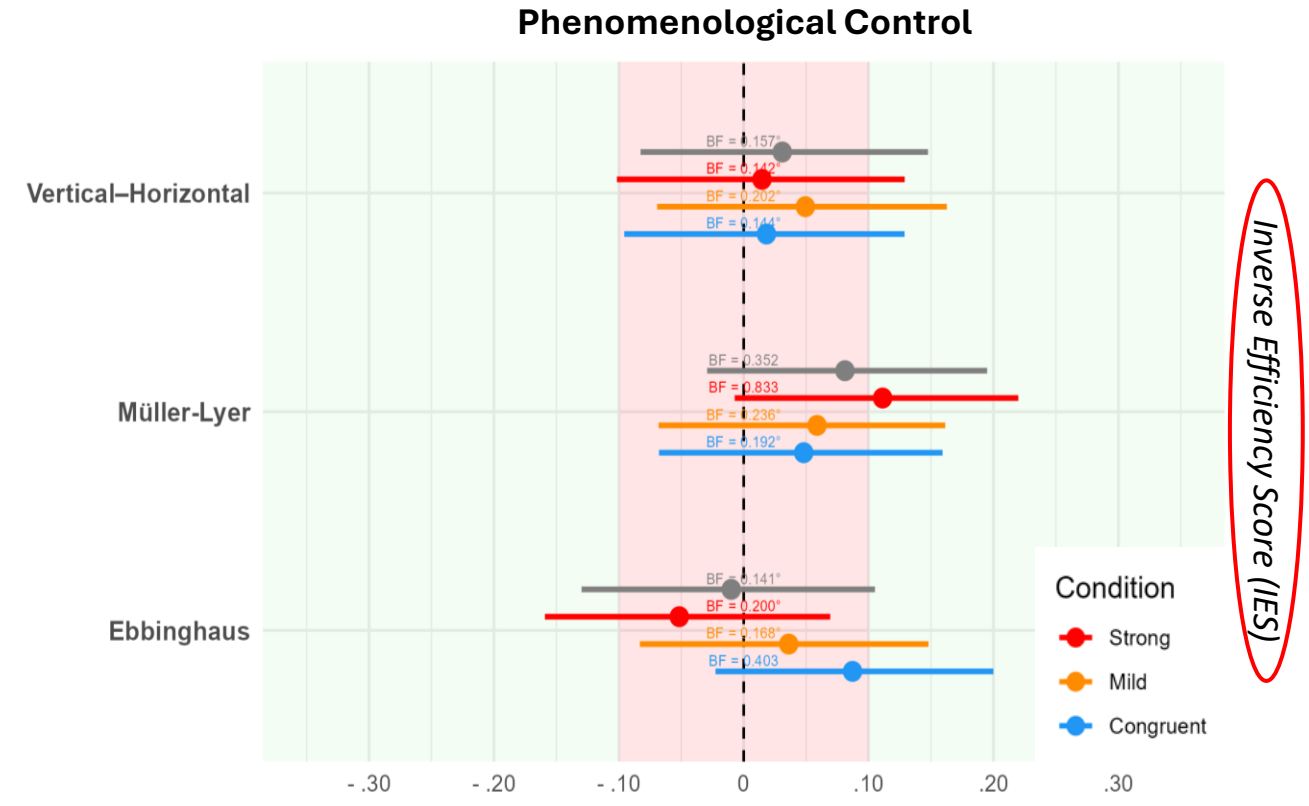
- Combines **speed** and **accuracy** responses into a single value
 - Adjusting RTs for errors
- Straightforward index of VI sensitivity
- Higher IES = more inefficient performance

$$\text{IES} = \frac{\text{Mean RT of correct responses}}{\text{Proportion of correct responses}}$$

Results

Did we replicate Lush et al. (2022)?

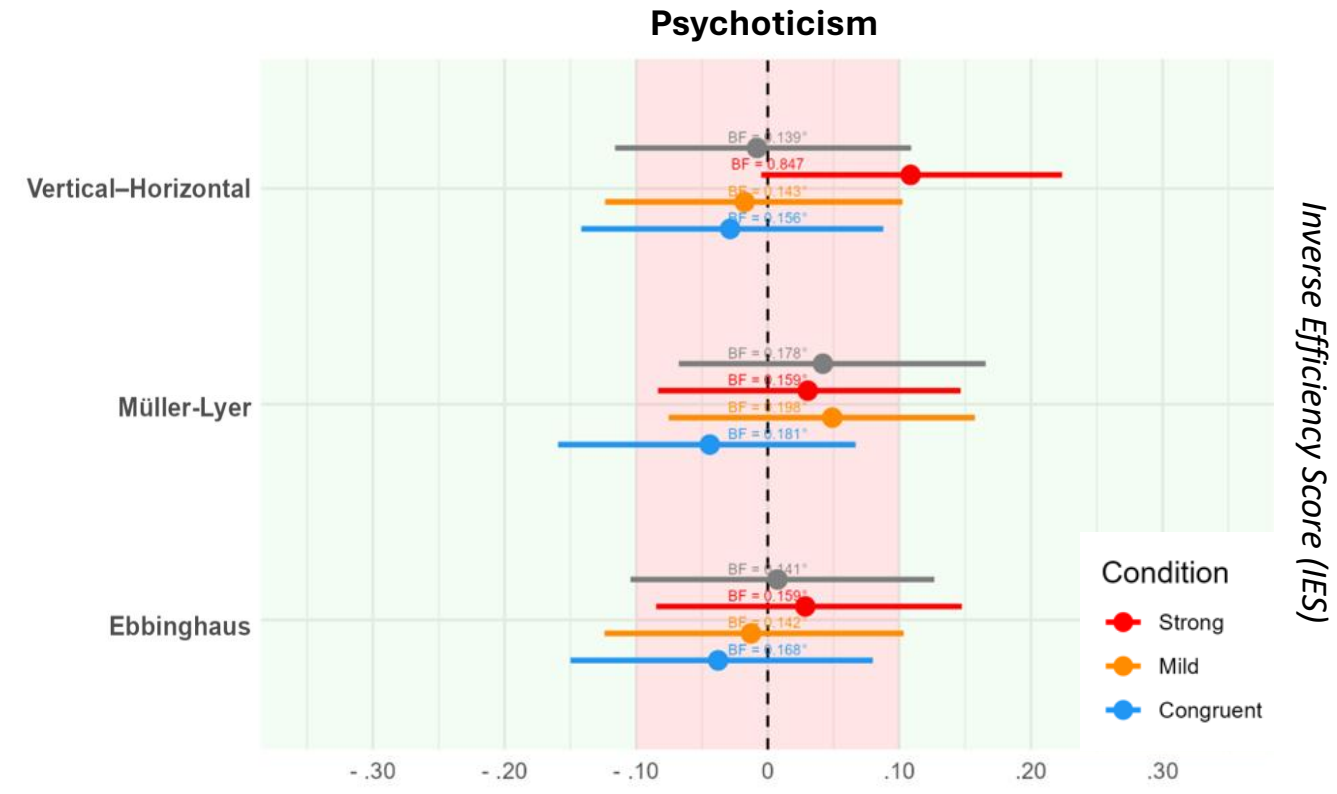
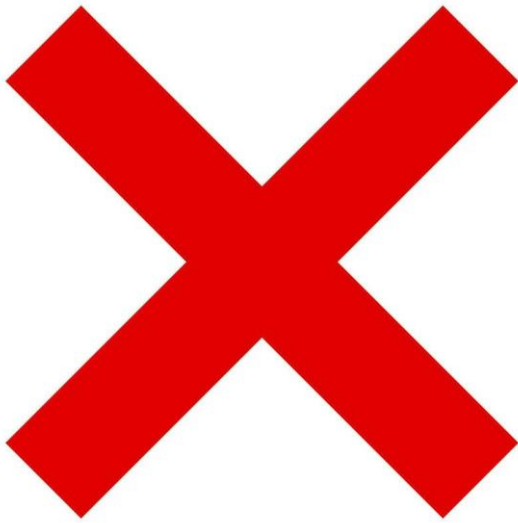
- Correlations yielded Bayes factors **less than 3**
- Overall, no clear pattern emerged



Results (Cont.)

Did we replicate Makowski et al. (2023)?

- All Bayes factors **less than 3**



Discussion

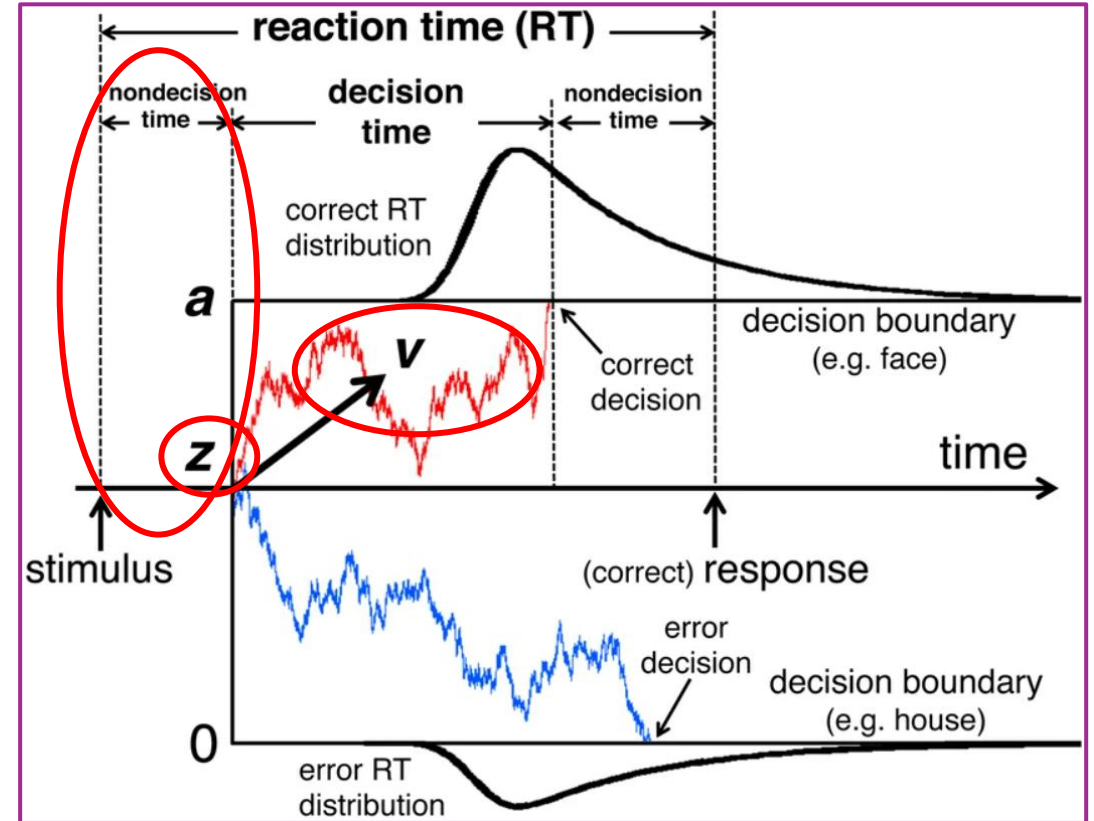
- Only first hypothesis was supported, replicating and extending Lush et al. (2022)
 - **Cognitive impenetrability** of visual illusions
- Thus, the ability to manipulate trait PC may be insufficient for altering perception of VIs
- Resistance to top-down effects

Why did we not replicate ourselves?

- Number of VI types employed
- Relatively crude measurement of VI sensitivity

Computational Cognitive Modelling

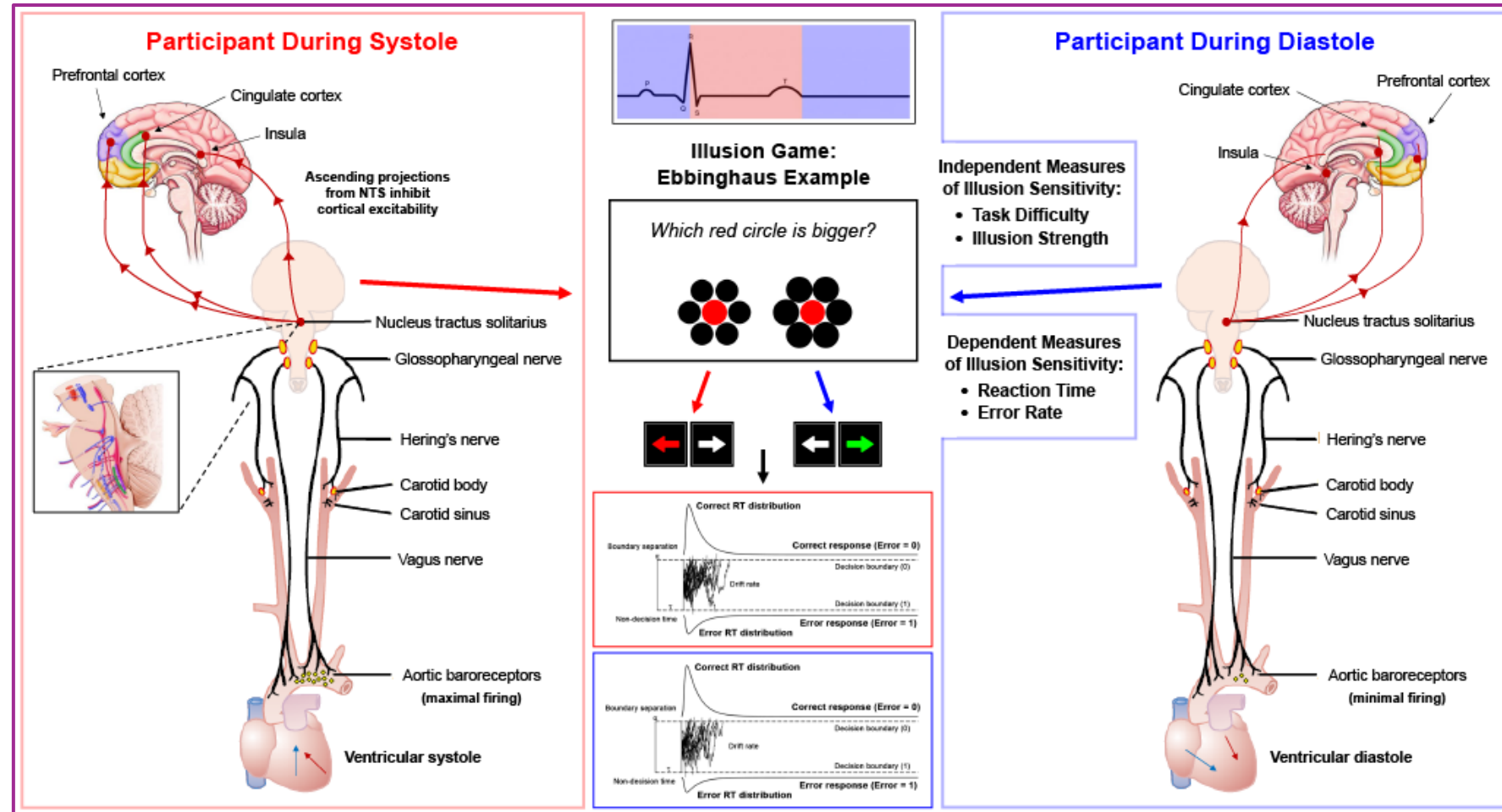
- Employing **evidence accumulation models** to IG data
- Ideal for modelling speed-accuracy data



Drift-diffusion modelling

Physiological Correlates of VI Sensitivity

My University of
Sussex **Junior**
Research Associate
(JRA) Scheme project!



Thank you for listening!